

AIR FRYER

MARKET

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DESCRIPTION OF AIR-FRYER MARKET

10

Name Brands
Analyzed

5

Years of
Data (2019–2023)

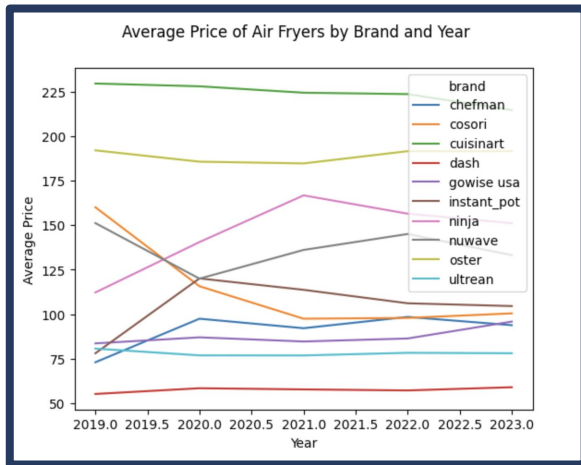
50

Brand-Year
Observations

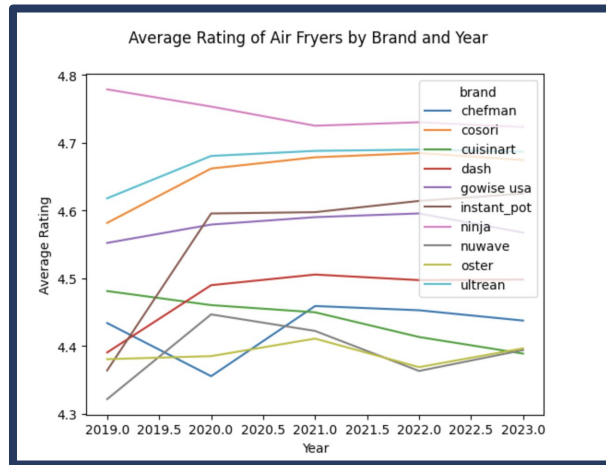
Key Findings

- GoWISE USA and Dash dominated in 2019 but have lost their standing by 2023.
- Instant Pot and Ninja are now market leaders and Chefman and Cosori rapidly growing.
- Cuisinart and Oster have the highest prices, however, Ninja consistently earns the highest consumer ratings.
- Prices and ratings have been stable and competition is mostly on brand identity and product features.

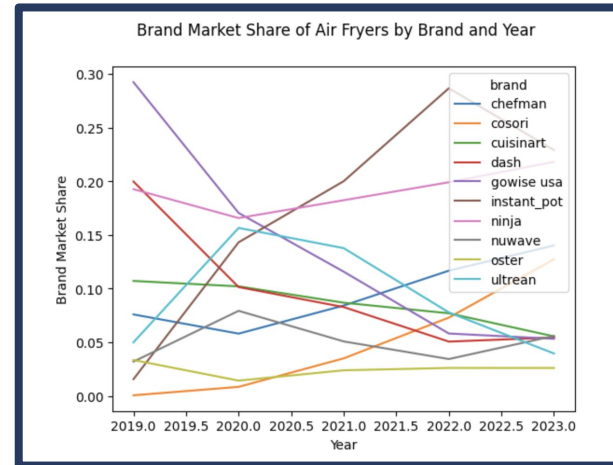
PRICES, RATINGS, AND MARKET SHARES OVER TIME



Cuisinart and Oster are the priciest brands; most others cluster below \$100 and stay flat over time



Ratings are tight across all brands (4.3–4.8); Ninja and Ultrean consistently lead.



GoWISE USA and Dash dominated in 2019 but collapsed; Instant Pot surged to the top by 2022–23.

THE DEMAND MODEL

What is a Logit Demand Model?

$$\log(s_{bt}) = \alpha_0 + \alpha_t + \gamma_b + \beta_{price}p_{bt} + \beta_{rating}r_{bt} + \sum_{\ell=1}^L \beta_{\ell}x_{bt\ell} + \epsilon_{bt}.$$

A logit-style demand model is used to understand how different factors affect a brand's market share. A brand's log market share relies on things such as price, quality, product features, brand identity, and year effects in this specific case. This is basically a multinomial logistic regression that we estimate using OLS on log shares. Year effects looks for how the overall market changes each year. Brand effects capture things about each brand that we are not able to directly measure in the data.

Model $R^2 = 0.763$

The model explains 76% of variation in log brand share.

Important Coefficients from Our Demand Model

Variable	Coeff.	Interpretation
Avg. Price	-0.038	Higher price, lower share
Avg. Rating	+0.288	Higher ratings boost share
Window share	+12.88	Strongest feature signal
Compact share	+9.82	Second strongest feature
Oven-style share	+1.94	Moderate positive
Rotisserie share	-5.67	Negative
Dual-basket share	-9.51	Very rare; negative

ESTIMATED PRICE COEFFICIENT & MOST IMPORTANT PRODUCT FEATURES

Price Coefficient

—0.038

per dollar increase in avg. price

Why the sign matters:

A negative price coefficient shows that consumers buy less at higher prices. If there was to be no negative price coefficient then the cost and markup estimates would fall apart.

	feature	coefficient
6	window_share	12.880298
2	compact_share	9.815304
8	brand_cuisinart	6.422436
12	brand_ninja	5.838705
11	brand_instant_pot	4.626260
10	brand_gowise usa	3.938996
14	brand_oster	3.928074
13	brand_nuwave	3.544883
7	brand_cosori	2.551946
4	oven_style_share	1.941774
15	brand_ultrean	0.942399
1	avg_rating	0.287517
9	brand_dash	0.176655
16	year_2020	0.119071
17	year_2021	0.041900
19	year_2023	-0.003307
0	avg_price	-0.037668
18	year_2022	-0.098860
5	roisserie_share	-5.674054
3	dual_basket_share	-9.509686

MOST IMPORTANT FEATURE:

Window Share

Coefficient: 12.88

SECOND MOST IMPORTANT FEATURE:

Compact Share

Coefficient: 9.82

Discussion of Most Important Features:

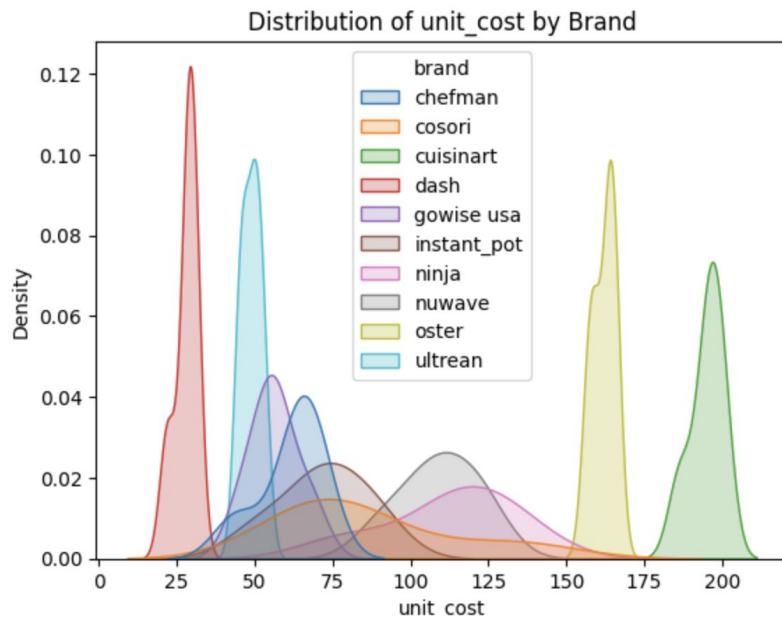
Consumers are more likely to buy air fryers with viewing windows and compact designs. Consumers value visible, practical product features the most. .

ESTIMATED UNIT COSTS, MARKUP, AND SHARE-WEIGHTED PROFIT

Estimated Unit Cost: dash \$27.94

Estimated Markup: ninja \$32.90

Estimated Share-Weighted Profit: ninja \$6.31



	unit_cost	markup	average_profit
brand			
chefman	61.569466	29.368946	2.820968
cosori	86.283321	27.984628	1.436651
cuisinart	194.896119	29.050974	2.502997
dash	27.936578	29.542349	2.994372
gowise usa	56.301040	31.153741	4.605764
instant_pot	71.903119	32.558234	6.010256
ninja	112.488470	32.854071	6.306093
nuwave	109.053549	27.970835	1.422858
oster	161.804969	27.226315	0.678338
ultrean	48.785765	29.328841	2.780864

Ninja shows the highest estimated markup and average profit, suggesting it has the strongest pricing power. Dash has the lowest inferred unit cost, which may reflect a lower-cost positioning strategy compared to higher-cost brands like Cuisinart and Oster.

BRAND POSITIONS & CONCLUSIONS

MARKET LEADERS

Instant Pot, Ninja

Highest share-weighted profit (\$6.31 and \$6.01). Ninja leads on ratings (4.74) and markups and Instant Pot on volume.

HIGHEST COST

Cuisinart, Oster

Highest unit costs (\$195 and \$162) but lowest average profits. High price does not correlate to consumer satisfaction.

RISING

GoWISE USA, Chefman

Mid-range costs with solid profit. GoWISE has the third highest average profit (\$4.61) even though it has a low price point.

FADING

Nuwave, Oster, and Cosori

Low average profits (\$0.68–\$1.44). Oster in particular has high costs but very low share-weighted returns.

SOURCES

Slideshow aesthetic sourced from Claude AI

Some code chunks sourced from Professor Terence Johnson's Github solutions